Name: Toppiiz ID: _____ No: _____

2110201 Quiz 2: Basic Skills

1. (0.5, 0.5, 2 points) Find the determinant of the following matrices

a. $\begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix}$

$$\begin{vmatrix} 1 & -2 \\ 3 & -4 \end{vmatrix} = 1(-4) - 3(-2) = -4 + 6 = 2$$

b. $\begin{bmatrix} 2 & -4 & 2 \\ 4 & -3 & 1 \\ 0 & -6 & 6 \end{bmatrix}$

$$\begin{vmatrix} 2 & -4 & 2 \\ 4 & -3 & 1 \\ 0 & -6 & 6 \end{vmatrix} = \left[2(-3)(6) + 4(-6)(2) + 0(-4)(1)\right] - \left[0(-3)(2) + 2(-6)(1) + 6(-4)(4)\right] = (-36-48) - (-12-96)$$

$$= -84 + 108 = 24$$

c. $\begin{bmatrix} 0 & 2 & -1 & 0 & 0 \\ 1 & 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & 4 & 1 \\ 1 & 0 & -1 & 0 & 5 \\ 0 & 0 & -3 & 1 & 1 \end{bmatrix}$

$$\begin{vmatrix} 0 & 2 & -1 & 0 & 0 \\ 1 & 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & 4 & 1 \\ 1 & 0 & -1 & 0 & 5 \\ 0 & 0 & -3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 2 & -1 & 0 & 0 \\ 1 & 0 & -1 & 1 & 2 \\ 0 & 0 & -3 & 1 & 1 \\ 0 & 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 4 & 1 \end{vmatrix} (-1) \quad ; \; -P_2 + P_4, \; P_3 \leftrightarrow P_5$$

$$= \begin{vmatrix} 1 & 0 & -1 & 1 & 2 \\ 0 & 2 & -1 & 0 & 0 \\ 0 & 0 & -3 & 1 & 1 \\ 0 & 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 0 & 13 \end{vmatrix} (-1)^2 \quad ; \; P_1 \leftrightarrow P_2, \; 4P_4 + P_5$$

$$= (1)(2)(-3)(-1)(13)\cancel{(-1)^2}$$

$$= 78$$

2. (1 point) Compute $AB + 2C$ where $A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & -3 & 2 \end{bmatrix}, B = \begin{bmatrix} 4 & -5 \\ -3 & 2 \\ 4 & -1 \end{bmatrix}, C = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$.

$$AB + 2C = \begin{bmatrix} 1 & -2 & 3 \\ 1 & -3 & 2 \end{bmatrix} \begin{bmatrix} 4 & -5 \\ -3 & 2 \\ 4 & -1 \end{bmatrix} + 2\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4+6+12 & -5-4-3 \\ 4+9+8 & -5-6-2 \end{bmatrix} + \begin{bmatrix} 2 & 6 \\ 4 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 22 & -12 \\ 21 & -13 \end{bmatrix} + \begin{bmatrix} 2 & 6 \\ 4 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 24 & -6 \\ 25 & -5 \end{bmatrix}$$

3. (3 points) Find a linear equation describing the hyperplane containing the points $\begin{bmatrix}2\\1\\2\\1\end{bmatrix}, \begin{bmatrix}4\\3\\2\\1\end{bmatrix}, \begin{bmatrix}1\\2\\3\\4\end{bmatrix}$, and $\begin{bmatrix}1\\1\\1\\1\end{bmatrix}$ in $\mathbb{R}^4$.

Equation: $\boxed{a x_1 + b x_2 + c x_3 + d x_4 = e}$

แทนค่า $\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}$ ด้วยจุดทั้ง 4

$$\begin{array}{ccccc} a & b & c & d & e \end{array}$$
$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & -1 & 0 \\ 2 & 1 & 2 & 1 & -1 & 0 \\ 1 & 2 & 3 & 4 & -1 & 0 \\ 4 & 3 & 2 & 1 & -1 & 0 \end{array}\right]$$

$$\xrightarrow[-4P_1+P_4]{-2P_1+P_2,\ -P_1+P_3} \left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & -1 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 & 0 \\ 0 & -1 & -2 & -3 & 3 & 0 \end{array}\right]$$

$$\xrightarrow[P_2+P_3,\ P_3+P_4]{P_2+P_1,\ -P_2} \left[\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \end{array}\right]$$

$$\xrightarrow[-\frac{1}{3}P_4+P_3,\ \frac{1}{3}P_4]{\frac{1}{3}P_4+P_2} \left[\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

$$\begin{array}{ccccc} a & b & c & d & e \end{array}$$
$$\xrightarrow[\frac{1}{2}P_3]{-\frac{1}{2}P_3+P_1} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

$\therefore \begin{bmatrix}a\\b\\c\\d\\e\end{bmatrix} \in \left\{ \begin{bmatrix}1\\-1\\-1\\1\\0\end{bmatrix} d \mid d \in \mathbb{R} \right\}$

Let $d = 1$;

$\boxed{x_1 - x_2 - x_3 + x_4 = 0}$ #

4. (1, 2 points) Find the inverse of the following matrices

a. $\begin{bmatrix} b & a \\ d & c \end{bmatrix}$

Case $bc - ad = 0$:

$\begin{bmatrix} b & a \\ d & c \end{bmatrix}$ is not invertible #

Case $bc - ad \neq 0$

$$\begin{bmatrix} b & a \\ d & c \end{bmatrix}^{-1} = \frac{1}{bc-ad}\begin{bmatrix} c & -a \\ -d & b \end{bmatrix}$$ #

b. $\begin{bmatrix} 3 & 1 & 0 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -1 & 0 & 1 \\ 2 & -1 & 0 & 1 \end{bmatrix}$

$\boxed{[A \mid I] \to [A^{-1}A \mid A^{-1}I] = [I \mid A^{-1}]}$

$$\left[\begin{array}{cccc|cccc} 3 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 2 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[-2P_3+P_4]{-3P_3+P_1,\ -2P_3+P_2} \left[\begin{array}{cccc|cccc} 0 & 4 & 0 & -3 & 1 & 0 & -3 & 0 \\ 0 & 3 & 1 & -1 & 0 & 1 & -2 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & -2 & 1 \end{array}\right]$$

$$\xrightarrow[P_4+P_3]{-4P_4+P_1,\ -3P_4+P_2} \left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 1 & 1 & 0 & 5 & -4 \\ 0 & 0 & 1 & 2 & 0 & 1 & 4 & -3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 0 & -2 & 1 \end{array}\right]$$

$$\xrightarrow[-2P_1+P_2,\ P_1+P_4]{P_1 \leftrightarrow P_3} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 & -2 & 1 & -6 & 5 \\ 0 & 0 & 0 & 1 & 1 & 0 & 5 & -4 \\ 0 & 1 & 0 & 0 & 1 & 0 & 3 & -3 \end{array}\right]$$

$$\xrightarrow{P_4 \leftrightarrow P_2} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 3 & -3 \\ 0 & 0 & 0 & 1 & 1 & 0 & 5 & -4 \\ 0 & 0 & 1 & 0 & -2 & 1 & -6 & 5 \end{array}\right]$$

$$\xrightarrow{P_4 \leftrightarrow P_3} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 3 & -3 \\ 0 & 0 & 1 & 0 & -2 & 1 & -6 & 5 \\ 0 & 0 & 0 & 1 & 1 & 0 & 5 & -4 \end{array}\right]$$

$$\therefore \begin{bmatrix} 3 & 1 & 0 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -1 & 0 & 1 \\ 2 & -1 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 0 & -1 & 1 \\ 1 & 0 & 3 & -3 \\ -2 & 1 & -6 & 5 \\ 1 & 0 & 5 & -4 \end{bmatrix}$$ #